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Operationalizing Accessibility: From Strategy to Practice

RESEARCH

ALI MOHAMED
JAN GULLIKSEN
SOFIA STRÖMQVIST
MASOOD RANGRAZ

*Author affiliations can be found in the back matter of this article

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ABSTRACT

Accessibility is increasingly framed as a rights and justice issue, yet many organizations still treat it as compliance rather than learning. This study examines how accessibility professionals conceptualize and operationalize accessibility across organizational contexts. Drawing on semi-structured interviews (n = 14) and one dyadic group discussion (n = 2), analyzed through reflexive thematic analysis, we identify four themes: empathy as organizational competence; policy, governance, and change management; technology, digital transformation, and collaboration; and capacity building for inclusive education and workplaces. The findings show that accessibility is operationalized not solely through technical standards but also through organizational learning, shared responsibility, and role-specific competence development. We contribute an empirically grounded, cross-sector account of how accessibility is translated from strategy into everyday practice, while delimiting the study to professional perspectives rather than compliance outcomes.

CORRESPONDING AUTHOR:
Ali Mohamed
Dalarna University, SE
moal@du.se

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1 INTRODUCTION

Across Europe, everyday participation depends on accessible products, places, and services in education, employment, transport, culture, tourism, and digital environments. Yet many people still cannot participate equitably. The EU Council infographic based on Eurostat data estimated that 27 percent of EU adults, about 101 million people, have a disability (European Council 2021). Directive 2019/882 has been in force since June 28, 2025, creating organization-wide obligations to demonstrate accessibility in practice (AccessibleEU 2025; European Union 2019). Despite mature frameworks, routine inclusion is not yet the norm. In the digital sphere, the WebAIM Million (2025) analysis of the home pages of the top one million websites reported a mean of 51 detectable errors per page, and 94.8% of pages with detectable issues mapped to WCAG 2.x success criteria, with persistent problems such as low-contrast text and missing alternative text (WebAIM 2025). 'Detectable' refers to issues found by automated checks; actual accessibility problems may be more extensive. WCAG 2.2 became a W3C Recommendation in 2023, (Henry 2023), and EN 301 549 v3.2.1 (ETSI et al. 2021) continues to harmonize requirements across the EU; yet standards alone have not produced consistently inclusive outcomes.

We address this knowledge-to-practice gap by examining how practitioners describe accessibility within the design, development, and service delivery processes. Drawing on interviews and discussions with three collaborating organizations, an academic partner (U1), a disability-advocacy organization (A1), and a social-innovation hub (I1), we surface shared challenges and situated solutions.

1.1 ACCESSIBILITY AS AN EVOLVING PRACTICE

Accessibility is both a rights- and justice-oriented commitment and a practical quality of sociotechnical systems. From a justice perspective, it interrogates disablism and ableism, seeking equitable participation in everyday life and treating exclusion not as an individual deficit but as a structural effect of norms, institutions, and designs (Goodley 2014; Tyler 2015). From an HCI practice perspective, accessibility is a quality attribute linked to usability, supported by methods, heuristics, and standards work that guide specification and evaluation (Gulliksen et al. 2003). Yet definitions and methods vary across health, education, public service, and industry, producing inconsistencies in how accessibility is scoped, measured, and maintained (Persson et al. 2015).

As digital systems mediate more of life, digital accessibility has become a core competence. Studies report capacity gaps and call for cross-role pedagogy that connects academic curricula with workplace practice, enabling practitioners to translate high-level norms into situated decisions (Coverdale et al. 2022; Lewthwaite et al. 2022). Operationalization increasingly depends on collaboration across roles and on shared design systems and knowledge platforms that embed accessibility in everyday development and content workflows (Hussain and Gulliksen 2024). Historically, many organizations have approached accessibility primarily as conformance, meeting criteria to pass audits, rather than as a commitment to lived inclusion and task success (Yesilada and Harper 2019). Disability-studies scholarship reframes the goal: removing barriers is necessary but insufficient unless the surrounding structures and responsibilities that normalize exclusion are also addressed (Goodley 2014).

Technology is an ambivalent mediator. Innovation can widen or narrow participation depending on governance, accountability, and whose assumptions are encoded into tools and services (Roulstone 2016). HCI work on dynamic diversity emphasizes that people's abilities vary over time and across contexts; interfaces and services should therefore accommodate variability rather than a single 'typical' user profile (Gregor et al. 2002). Taken together, these strands shift the design problem away from one-off fixes toward systems and organizations capable of continuous adaptation.

Against this backdrop, our study examines how professionals responsible for accessibility define, prioritize, and operationalize it in daily work. We analyze how they coordinate across roles, balance standards with context, and navigate tensions between rights-oriented commitments and technology-led change while striving for outcomes that matter to users (Coverdale et al. 2022; Goodley 2014; Gregor et al. 2002; Gulliksen et al. 2003; Hussain and Gulliksen 2024; Lewthwaite et al. 2022; Persson et al. 2015; Tyler 2015; Yesilada and Harper 2019). In the next section, we outline how this applies to digital products and services.

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1.2 DIGITAL ACCESSIBILITY AND EVOLVING FRAMEWORKS

Digitalization mediates learning, work, culture, and civic participation. Accessibility, therefore, encompasses not only the built environment but also the design and governance of websites, apps, and data-driven and AI-enabled systems (Ara et al. 2024). In this article, however, we focus analytically on digital accessibility and on the organizational practices through which it is implemented. References to cultural or built-environment contexts are retained only where participants used them to illustrate cross-sector organizational challenges. In this domain, organizations navigate three complementary instruments: (a) legislation (e.g., Directive (EU) 2019/882), (b) standards (e.g., EN 301 549 specifying harmonized ICT requirements), and (c) technical guidelines (e.g., WCAG, a W3C Recommendation widely adopted for web and software accessibility) (ETSI et al. 2021; Gulliksen et al. 2003; Gulliksen et al. 2004; Henry 2023). Together, these provide baselines and shared reference points for planning, procurement, and evaluation; yet robust accessibility still relies on iterative design, authentic user involvement (including assistive-technology use), and organizational cultures of empathy, accountability, and continuous learning.

1.3 RESEARCH PROBLEM AND GAP

Despite advances in technical standards, evaluation methods, and user experience guidance, empirical insight remains limited into how accessibility professionals in education, public services, cultural heritage, and digital design conceptualize, implement, and advocate for digital accessibility in everyday organizational work across sectors. We know little about how teams make situated decisions, collaborate across roles, exercise leadership, and learn to move beyond checklists and late-stage fixes. Adjacent service contexts are considered only to the extent that they illuminate how accessibility is translated into organizational routines, responsibilities, and learning practices. The field still lacks understanding of four issues: how empathy is cultivated as a professional competency; how WCAG and the EAA become project-level definitions of done and acceptance criteria; how cross-functional collaboration and user participation occur under real-world constraints; and which emerging technologies, including AI, are adopted, and how they are adopted critically and ethically. We identify a gap at the intersection of digital accessibility, organizational practice, and learning, and call for research on how professionals co-construct accessibility as a learning-oriented, culturally embedded practice that supports sustained, user-centered transformation and social sustainability.

1.4 AIM AND RESEARCH QUESTION

We examine how accessibility professionals conceptualize, implement, and advocate for accessibility within organizational contexts, with a focus on digital products and services. Accordingly, we ask: How do professionals responsible for accessibility conceptualize, implement, and advocate for accessibility within their organizations?

1.5 THEORETICAL ORIENTATION: A SOCIOCULTURAL PERSPECTIVE ON LEARNING

We interpret the results through a sociocultural perspective, in which learning is understood as participation in socially organized activity mediated by cultural tools and artifacts. By mediating artifacts, we refer to material and symbolic tools that shape and extend action in practice, such as standards, templates, checklists, design systems, prototyping software, and evaluation procedures. Such artifacts carry histories of use and facilitate coordinated work. Language operates as the central mediating tool, a flexible sign system that supports description, coordination, reflection, and the accumulation of shared social memory. Through language and other external records, organizations build a shared social memory, that is, knowledge stored in documents, repositories, and routines that a group can consult over time, such as wikis, knowledge portals, decision logs, and code or design libraries. Capability grows through guided participation supported by scaffolds. By scaffolds, we mean temporary and adaptive supports; such as those from colleagues, mentors, or tools, that structure participation until competence increases. Scaffolds can be human, such as pairing or coaching, or artifact-based, such as stepwise checklists and templates; as people gain mastery, the supports are gradually removed (Hillman and Säljö 2016; Säljö 2016). Read in this way, standards and laws (for

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example, WCAG, EN 301 549, and Directive (EU) 2019/882, checklists, prototypes, and meeting rituals operate as mediating artifacts, while peer learning, mentoring, and knowledge portals make social memory actionable by converting values into durable routines. We draw on this orientation in the Discussion to explain how organizations learn their way from compliance toward sustained inclusion. We revisit these constructs in Section 4 to interpret our findings. We contribute a practical account of accessibility as organizational learning, showing how specific governance roles, acceptance criteria, and indicators translate the European Accessibility Act (EAA) into day-to-day technology and collaboration. Our analysis focuses on how professionals make meaning of and operationalize accessibility in organizational practice; it does not evaluate national compliance outcomes or provide an audit of specific services.

2 METHOD

Analyzing professionals' experiences reveals how accessibility principles are implemented or hindered in real-world settings. It indicates whether organizational cultures treat inclusion as a robust ethic supported by empathy and iterative improvement, or as a narrow compliance task. Using reflexive thematic analysis (RTA), (Braun and Clarke 2006; Braun and Clarke 2019; Braun and Clarke 2021), this study captures how accessibility is perceived, operationalized, and sustained across sectors, offering an account of accessibility as a culturally embedded practice that supports user-centered inclusion.

2.1 DESIGN AND ANALYTIC APPROACH

This study employed RTA to investigate how professionals conceptualize, implement, and advocate for accessibility across various sectors. Grounded in a constructionist stance, we treated accounts as situated narratives and acknowledged our active interpretive role. The analysis considered organizational culture, cross-functional collaboration, user participation, and project constraints alongside technical conformance. We maintained reflexive memos, kept an audit trail of analytic decisions, and held regular dialogues with a critical friend to challenge our assumptions. We treated coding as interpretive and supported credibility through reflexivity, team discussions, and a documented audit trail. By reflexivity, we mean the ongoing self-scrutiny of how our positions and values shape the analysis; by theoretical transparency, we mean the explicit reporting of those assumptions and choices; and by interpretive depth, we suggest developing themes that capture patterned meaning rather than providing topic summaries (Braun and Clarke 2006; Braun and Clarke 2019; Braun and Clarke 2021). We interpreted patterns through a sociocultural lens in which artifacts mediate activity, practices are locally appropriated, and scaffolds support guided participation (Säljö 2016; Wenger 1998).

2.2 DATA COLLECTION

We collected data via semi-structured interviews (n = 14) and one dyadic group discussion (n = 2) with participants across roles and seniority levels (N = 16). Using purposive sampling, we sought variation across sectors, roles, and levels of involvement, intentionally recruiting participants relevant to the research questions rather than drawing a random sample. Interviews were conducted either online or in person, depending on the participant's preference and location. Sessions lasted approximately 45–90 minutes and followed a guide that covered roles, experiences of implementing accessibility, perceived barriers, collaboration and training practices, and organizational culture. To protect confidentiality, collaborating organizations are anonymized as an academic partner (U1), an advocacy organization (A1), and an innovation hub (I1). For U1, sessions were audio- or video-recorded and transcribed verbatim. For A1, interviewer-prepared summaries were produced rather than recordings or verbatim transcripts. For these summaries, interviewers produced structured write-ups capturing key points, examples, and (where feasible) short verbatim phrases; in analysis, we treated them as lower-granularity material and used them primarily for contextual triangulation rather than as primary evidence for fine-grained interpretive claims. For I1, both recordings and verbatim transcripts were available. All transcripts and summaries were anonymized; identifying details were replaced with pseudonyms or role-based descriptors.

Dyadic group discussion rationale and analytic handling. The two-participant session involved interaction between participants (building on and responding to each other's accounts) and

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was therefore treated as a dyadic group discussion rather than a standard one-to-one interview. We analyzed it using the same reflexive thematic analysis procedures as the interviews, while acknowledging that a dyadic format offers less breadth of group dynamics than larger focus groups.

2.3 ETHICAL CONSIDERATIONS

The study followed ethical guidelines for research with human participants and received approval from the relevant ethics review board before data collection. All participants provided informed consent, in writing or verbally depending on interview format, after receiving information about the study's purpose, procedures, voluntary participation, and data protection. Data security measures included storing recordings and transcripts on encrypted, password-protected drives accessible only to the research team, removing personal identifiers from all outputs, and honoring withdrawal requests by deleting data. In line with (SRC 2024) guidance, we collected only the information necessary for the study objectives and will securely destroy all personal data upon project completion.

2.4 PARTICIPANTS

A total of 16 professionals participated, including one dyadic group discussion with two participants. Participants had varying levels of involvement in accessibility, ranging from formal responsibilities to advocacy or adjacent roles. The sample spanned sectors and contexts: education (university lecturers, educational technologists, special needs coordinators); public sector and policy (accessibility consultants, digital inclusion specialists, city planners); cultural and heritage institutions (museum educators, library service managers); healthcare and rehabilitation (therapists, project managers in health-related accessibility projects); and technology and digital design (UX designers, developers, digital service providers). Experience ranged from more than fifteen years to early-career entrants from education, communications, or human-computer interaction. Some held formal training in WCAG and EN 301 549; others developed expertise through practice, community engagement, or collaboration with user groups. This heterogeneity provided multiple perspectives on organizational positions, resources, and pathways into accessibility work.

2.5 ANALYSIS

Following reflexive thematic analysis (RTA), we iteratively moved through six phases (Braun and Clarke 2006; Braun and Clarke 2019; Braun and Clarke 2021). First, we familiarized ourselves with the corpus by reading and rereading transcripts, listening to audio, and writing memos capturing first impressions and context. We coded interview transcripts and the dyadic discussion transcript in full; interviewer-prepared summaries were coded with attention to their lower level of detail and were used to support, not drive, theme development. Second, we generated initial codes across the dataset. Salient features included empathy, legal frameworks, digital transformation, user testing, education, leadership, and implementation challenges. Third, we clustered related codes and mapped relations across sectors, retaining productive overlaps (e.g., links spanning education and employment). Fourth, we reviewed themes for coherence and distinctiveness against coded extracts and the full dataset. Fifth, we defined and named themes, clarifying scope, claims, and subthemes. Sixth, we produced an analytic narrative that integrated themes, illustrative quotations, and implications for practice and future directions. We treat themes as analytic outputs rather than entities 'discovered' in data, consistent with RTA. The sample size supported close engagement with each transcript and the development of rich, well-supported themes.

3 RESULTS

We identified four themes and assigned compact tags for cross-reference: 3.1 Empathy as organizational competence (T1), 3.2 Policy, governance, and change management (T2), 3.3 Technology, digital transformation, and collaboration (T3), 3.4 Capacity building for inclusive education and workplaces (T4).

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3.1 EMPATHY AS ORGANIZATIONAL COMPETENCE (T1)

Across education, technology, public services, business, and cultural heritage, participants described effective accessibility as standards-informed and relational, grounded in ethical commitment, empathy, and cultural competence. Rather than a matter of regulatory compliance alone, they framed it as a commitment to fairness and participation, informed by WCAG, Directive (EU) 2019/882, and EN 301 549, and realized through everyday choices, interactions, and design decisions. This opening theme integrates findings on empathy and user experience, cultural competence and ethical leadership, and access in cultural and tourism settings, including pragmatic, budget-conscious solutions. Although participants referred to several adjacent service domains, we interpret these examples here as instances of organizational accessibility practice rather than as separate domain-specific analyses. We treat empathy here as an emergent theme from participants' accounts and connect it to prior scholarship in the Discussion. The inductive emphasis on empathy aligns with human-centered design scholarship that treats empathy as an organizational competence operationalized through research methods, curated user evidence, and sustained stakeholder engagement rather than sentiment (Drouet et al. 2023). Accessibility pedagogy likewise frames empathy as a shared, cross-role capacity developed via role-sensitive and mixed-role training, clearer cross-disciplinary communication, and 'champions' networks that keep practice connected (Coverdale et al. 2022; Lewthwaite et al. 2022). These connections help explain participants' reports that scenario-based activities, direct user involvement, and routine feedback loops become durable parts of everyday work.

Empathy and user experience: Participants consistently emphasized experiential learning, scenario-based activities, role-play, and guided walkthroughs of common barriers as catalysts that move teams from abstract sympathy to practical empathy. These activities were described as effective for surfacing specific obstacles and informing design decisions. Two interviewees stressed the value of tailoring learning materials and software interfaces to particular user needs. One illustrated how a seemingly minor detail became exclusionary in a computerized test: the choice of letters introduced confusion and blocked progress. We note that impairment simulations are contested in the literature and may risk trivializing lived experience or giving false confidence. When used, participants reported pairing them with structured reflection, direct involvement of disabled people, and concrete design changes. The Discussion returns to these tensions and outlines safeguards and alternatives, including co-design with organizations of persons with disabilities (OPDs), user panels, and evidence from real usage: 'It is important to be able to customize. I had a computer-based test and wondered whether there were particular letters I should avoid. B and D look similar. It can matter; it becomes an unnecessary barrier' (P11). 'For my part, where I am now, it is changing management; how we move from why to how, empathy, not sympathy. We need to talk about this much more,' (P11). 'Working with people is an absolute requirement' (P15).

Multiple sources of evidence across interviews and analytic memos converged on the value of perspective-taking activities, such as guided walkthroughs and role-play (e.g., dyslexia overlays, color-blindness filters, keyboard-only use). Participants also stressed that disability can be dynamic or invisible, which argues for continuous awareness so that empathy becomes habitual rather than occasional. In short, they prioritized accessibility as a human practice in which small, concrete choices about typography, instructions, and affordances are checked against lived experience and user evidence from co-design, task-based testing with assistive technologies, analytics, support tickets, and feedback channels.

To anchor how empathy travels into day-to-day work, participants described co-planning with students and colleagues to tailor strategies to individual needs, emphasizing day-to-day adjustments guided by what works in practice (P4). Several linked empathies to norm-aware listening and non-labeling classroom relations, noting that students often articulate concrete needs when instructors listen carefully and adjust routines accordingly (P7).

Cultural competence and ethical leadership: Participants argued that these are indispensable complements to technical skill. Leaders who champion accessibility beyond compliance help shape the organizational culture required for sustained practice. Several accounts echoed the view that accessibility is a matter of cultural competence and ethics, and that leadership must normalize questions such as: whose participation is at stake, and what does inclusion look like in this context?

Cultural, heritage, and tourism accessibility: Participants extended the foundational mindset into the cultural and heritage domain, repeatedly invoking democracy and the right to cultural participation: 'Cultural heritage is a matter of democracy. Everyone should be able to take part in our shared cultural heritage, regardless of ability or background' (P5). We retain these examples as illustrative extensions of organizational accessibility work across service contexts, rather than as a shift in the article's core analytical focus. Where physical access is limited, participants proposed technological solutions, including VR/AR tours and digital archives, while emphasizing that these tools must themselves be accessible (for example, text alternatives and sign-language overlays): 'It may not be possible always to visit a physical site, but it should be made accessible by other means. Technology might be useful' (FG-2P).

Beyond ramps or Braille signage, participants emphasized staff readiness to guide visitors with hearing or cognitive impairments. They recommended short, adaptive tour-guidance modules to help staff adjust language, pacing, and demonstration methods, as well as practical micro-interventions with a significant impact on participation.

Tourism and hospitality, 'welcome all guests': In tourism and hospitality, many frontline teams were perceived as underprepared for diverse disabilities, particularly those beyond mobility: 'Hospitality facilities must be adapted to guests with wheelchairs, but there is no knowledge on how to approach people with other kinds of disabilities. Maybe they do not feel it is a space for them' (P2). Participants framed mandatory 'welcome all guests' training as both an ethical imperative and a business advantage. They also recommended self-audit toolkits (e.g., checklists, user journeys) to expose hidden pitfalls, such as confusing signage or inaccessible kiosks: 'To be able to inform tourist organizations on how good accessibility is a huge opportunity' (P5).

Budget-conscious, practical solutions: Budget shortfalls, common in cultural and nonprofit institutions, were not seen as grounds for delaying action. Participants advocated incremental, user-tested improvements (for example, tactile signage, audio guides, wayfinding, user-tested websites) and suggested pooling resources, sharing templates, or seeking local grants and philanthropic funds. Even where large-scale renovations are out of reach, low-cost steps can meaningfully improve access now, particularly when driven by direct user input.

Cross-cutting mindset: While empathy, cultural awareness, and leadership can be parsed as subtopics, participants consistently presented them as one holistic inclusion mindset, a normative, ethical, and cultural orientation rather than a checklist. This mindset permeates subsequent themes, anchoring discussions about governance, collaboration, and continuous improvement. Having established the ethical and cultural foundations, participants repeatedly asked, 'How do we make this stick?' The next theme shows how policy, governance, and change management (T2) translates values into responsibility, processes, and accountability.

3.2 POLICY, GOVERNANCE, AND CHANGE MANAGEMENT (T2)

Participants highlighted a persistent tension between compliance-driven and user-driven approaches. Standards and legislation, WCAG, Directive (EU) 2019/882, and EN 301 549 are essential baselines and motivators, but they do not guarantee meaningful inclusion on their own. Professionals described how governance, leadership, and transparent chains of responsibility determine whether accessibility remains a frontstage promise or becomes a backstage reality in day-to-day work.

Legal and policy awareness, from floor to practice: Participants called for explicit operational criteria, often phrased as a Definition of Done, to bridge the gap between regulatory text and practical delivery: 'The democratic right to information. Everyone should be able to access public administration. It should not be a post-construction' (P8). 'We are working extensively with the regulations around what is coming with Directive (EU) 2019/882, and how we will handle it at the agency level' (P8). 'There are two parts to requirements work: what you set before the work starts, and then the Definition of Done' (P6).

Participants repeatedly characterized law as the floor, not the ceiling, and stressed translation into milestones, acceptance criteria, and quality gates tied to user outcomes. Staff asked for clear, school-wide, and organization-wide guidance, so all roles, not only specialists, know how to communicate with and support people with disabilities (P4, P9). Respondents also framed

universal design as a baseline duty in public services, tied to democratic rights to information and participation, rather than an optional add-on (P12). Several emphasized legal literacies across the organization so policy can be translated into practical decisions at case level (P14).

Leadership and organizational culture: Change management emerged as decisive. Leadership engagement, allocating resources, modeling inclusive behavior, and requiring training turned accessibility into a shared responsibility rather than an isolated task. With effective leadership, organizations were more likely to integrate accessibility into procurement, IT, communications, facilities, and HR routines.

Responsibility chains, metrics, and accountability: A common barrier was uncertainty over 'who owns accessibility.' Participants proposed formal governance, such as accessibility steering groups, to prevent issues from falling through organizational cracks. They advocated tracking and reporting mechanisms, for example, a small set of KPIs such as 'percent of new digital assets passing WCAG 2.2' to make progress visible and hold teams accountable. Normalizing experimentation, and even failure, was seen as essential to learning: 'I wish I knew much more about how people function, about different disabilities, and that we could fail along the way and talk about it' (P8). Policy and governance set the rules of the game, but participants also stressed the practicalities of implementation, especially where technology and logistics intersect with collaboration and user feedback. The next theme addresses that operational layer.

3.3 TECHNOLOGY, DIGITAL TRANSFORMATION, AND COLLABORATION (T3)

Participants navigated a rapidly digitizing landscape, including AI-enabled systems, new platforms, and evolving user expectations, while warning against technology-first approaches that overlook fundamental usability and accessibility. They championed cross-functional collaboration, iterative testing, and logistical readiness, including safety, as the mechanisms that make accessibility work in everyday projects. For adult distance learners, collaboration with municipal study centers provides a physical place and staff contact that offsets the absence of on-site support; teams should agree on what can be handled remotely and what requires local assistance (P3). 'Being reachable by email or phone is even more important at a distance; students should not feel forgotten' (P3).

AI and technology promise, with precautions: Participants urged cautious integration of AI. Without foundational accessibility literacy, AI risks embedding biases or overshadowing essential usability. 'We have a huge challenge ahead of us. What should our AI be based on?' (P6). Another participant highlighted overlooked everyday functionality when teams chase advanced features: 'Sometimes my students tend to think about function, not the simple stuff like how to put a device on. That is also part of accessibility, especially if AI is involved in controlling it' (P13). Participants also reported situational benefits of AI for reading and reflection alongside concerns about ableist and other biases, arguing for human review and bias-mitigation layers (P7). Service design, ethics, and AI literacy were seen as topics that must be integrated into role-specific training for managers, agile coaches, and development staff rather than treated as stand-alone specialties.

Collaborative digital workflows and continuous improvement: The best outcomes were reported when accessibility was integrated from project kickoff, with checkpoints embedded in agile ceremonies and tools. This approach reduces late-stage retrofits and signals priority. User testing and feedback loops, including peer-led labs, were valued even in low-budget contexts because they keep teams focused on concrete barriers and real user experience. In this sense, collaboration functions as a learning mechanism: teams collectively define the Definition of Done and sharpen it over time.

Infrastructure, logistics, and safety (emergency preparedness): Participants rarely saw emergency preparedness addressed in standard accessibility work but considered it an important, situated example of operational accessibility where policy, facilities, and lived experience intersect. Buildings, especially older ones, require inclusive evacuation planning, clear signage, and route design that accounts for diverse mobility and sensory needs: 'There needs to be better solutions for evacuating, for example, people in wheelchairs. We need to be trained in scenarios. Older buildings can be tricky,' (FG-2P).

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Suggested actions included inclusive drills and cross-training facilities staff so that safety becomes part of universal design rather than an afterthought. These logistical practices are not peripheral; they are where policy, technology, and human factors converge.

Implementation is ultimately sustained by people who learn, teach, hire, mentor, and iterate together. The final theme gathers participants' accounts of education, workplace inclusion, and long-term capacity building.

3.4 CAPACITY BUILDING FOR INCLUSIVE EDUCATION AND WORKPLACES (T4)

Participants repeatedly returned to training and learning, not as one-off workshops but as ongoing, distributed practices that shape how organizations define and prioritize accessibility. This theme consolidates findings on education, employment inclusion, knowledge infrastructures, and leadership-supported change.

Inclusive education and pedagogy (cross-role learning): Although sector contexts varied, educational recommendations converged on a few key points: train broadly, including leaders; focus on Universal Design for Learning (UDL) (Izzo and Bauer 2015); and integrate experiential empathy-building activities such as scenario-based exercises and real user engagement. Micro-credentials and mentorships were described as ways to keep educator learning continuous, recognized, and shared. Teaching for diversity requires institutional support so it does not depend on individual teachers' goodwill. Educators asked for clear, highly structured instructions, for example, numbered steps and explicit expectations, which help students with ADHD and also benefit everyone (P7). They emphasized adapting materials and pacing to different reading and processing speeds rather than expecting uniform uptake (P4). Participants asked for practice-oriented exercises, not abstract lectures, so that teams can try methods and immediately apply them in their own settings (P1). Several highlighted the value of a basic, working understanding of common diagnoses and plain, highly structured communication of instructions and feedback (P3).

Workplace accessibility and employment inclusion: Participants emphasized accessible hiring platforms, explicit job materials, and onboarding 'buddies' trained in disability awareness, since these interventions influence who applies, who is hired, and how quickly new employees can contribute. Training for hiring managers and HR was cited as particularly impactful, making inclusion visible in upstream processes rather than treating it as post-hire accommodation. Public sector employers were encouraged to audit e-recruitment platforms and onboarding routines for accessibility gaps.

Train-the-trainer systems and knowledge portals: Participants endorsed train-the-trainer strategies to distribute expertise internally. Rather than relying on a small number of champions, organizations can pair early adopters with department heads and mentors who cascade skills across teams. Complementing this, many advocated for curated knowledge portals as a reliable source of truth featuring checklists, code examples, and success stories, to help practitioners trust the information they find online and avoid misinformation: 'We need a reliable source of knowledge, so I can know if resources online are true. That happens a lot in the accessibility environment,' (P10). Several emphasized stepping out of comfort zones and doing tasks oneself, to experience barriers first-hand: 'People need to step out of their comfort zone and do it themselves sometimes, to realize how it is. That helps shape practical and empathic solutions' (P13).

Holistic training and change management: P6 underscored that future training programs must be broad in scope and role-differentiated, and argued for researching 'non-users,' people who are not yet engaging due to barriers, to uncover latent issues: 'It is extremely important. I often talk about researching the non-user, people who are not engaging because of barriers' (P6).

This stance aligns with other participants' calls for top-down support, clear guidelines, and ongoing mentorship. Using a simple frontstage and backstage metaphor, public commitments to accessibility work only when backstage processes, including budgeting, staffing, procurement, and governance, are coordinated. When leaders fund learning and assign ownership, organizations turn intentions into sustained routines. Frontline staff training should include professional conversation methods and boundary-setting, along with basic knowledge of disability concepts and applicable regulations, to ensure support is effective and sustainable (P14).

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Next steps and continuous evolution: Participants outlined a pragmatic path: keep training iterative, make user engagement routine, invest in knowledge hubs, and align policy milestones with practical Definition of Done criteria. Accessibility is an evolving practice, a cycle of learning, testing, measuring, and refining rather than a one-time achievement.

4 DISCUSSION

This study contributes beyond synthesis in three ways. First, it shows across sectors how accessibility is enacted as organizational learning through shared tools, guided participation, and social memory, rather than merely as technical compliance. Second, it refines prior accessibility research by tracing how empathy becomes operational competence through governance routines, role-specific training, and accountability structures. Third, it extends existing discussions of implementation by showing how standards, collaboration, and organizational logistics interact in day-to-day practice.

Table 1 presents a theme-based synthesis of cross-cutting training needs reported by participants across sectors. The table is intended as an analytical condensation of recurring issues across participants' accounts in the interviews and dyadic discussion, not as a frequency count or a stand-alone framework. Each row summarizes patterns discussed in Themes T1–T4 and is grounded in the empirical material presented in the Results. It is not offered as a stand-alone prescriptive checklist; rather, it maps recurring competencies back to the four empirical themes and illustrates how practitioners translate values and legal requirements into learning, routines, and accountability, cross-cutting training needs aligned with the four themes. Empathy and user-centered design are foundational. Participants emphasized barrier-focused, user-led activities and direct engagement with people with disabilities to move from abstract sympathy to practical understanding. Some used role-play or impairment simulations for initial awareness, but these practices are contested and, if used, should be co-designed, critically examined, and never replace user involvement. This emphasis aligns with a sociocultural view on learning in which understanding is co-constructed through participation, shared tools, and situated interaction. In our material, empathy serves as a design and decision-making resource, shaping typography, interaction flows, and service scripts. Immersive workshops and real user interviews make it routine in everyday work.

A sociocultural lens explains why these patterns recur. Accessibility strengthens when mediating artifacts are usable in daily work, when scaffolds support guided participation, and when knowledge is stored in organizational social memory (Säljö 2016). In direct response to the research question, professionals conceptualize accessibility as justice-oriented and standards-informed; they implement it through governance that translates law and standards into acceptance criteria and a clear Definition of Done; and they advocate for it by cultivating empathy and cultural competence, securing leadership sponsorship, and building routines that make inclusion durable.

These patterns align with disability studies that treat exclusion as socially constructed and changeable through institutional practice, governance, and design (Goodley et al. 2019). They also resonate with Human-Computer Interaction accounts of dynamic diversity that argue variability should be designed for from the outset (Gregor et al. 2002). Evidence on shared design systems and cross-stakeholder collaboration shows how standards are operationalized (Hussain and Gulliksen 2024). Work on empathy and cross-role pedagogy explains how competence is built and sustained (Coverdale et al. 2022; Drouet et al. 2023; Lewthwaite et al. 2022).

Recent empirical studies strengthen this interpretation. Research with accessibility practitioners highlights persistent gaps in training, tensions between formal compliance and user-centered evaluation, and the importance of organizational capacity in accessibility work (Seixas Pereira and Duarte 2025). Other empirical work suggests that technical standards alone are insufficient unless complemented by user feedback and attention to the intelligibility of information (Baudet et al. 2024), while audit-based studies of higher education websites indicate that legal requirements matter but do not automatically produce accessible outcomes (Laamanen et al. 2024). Against this background, the present study extends recent empirical research by showing how accessibility is operationalized through organizational learning, collaboration, and accountability across sectors.

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TRAINING CATEGORY	CORE COMPETENCIES	THEME	KEY POINT AND IMPLEMENTATIONS
Empathy and user-centered design.	Scenario role-play, interviews with users with disabilities.	T1	Center lived experience, use barrier-focused methods, and favor immersive workshops over simulations.
Legal and policy awareness.	WCAG 2.2, Directive (EU) 2019/882, EN 301 549, local standards.	T2	Clarify EU–local overlaps, consider formal certification to professionalize practice.
AI and emerging technologies.	AI ethics, bias mitigation, practical tool literacy, and dataset provenance.	T3	Live demos show AI limits, prioritize hands-on sessions before deployment.
Digital transformation and collaboration.	Cross-functional training, partnerships, agile accessibility checkpoints.	T3	Start at project kickoff, blend integrated and stand-alone formats.
User testing and continuous improvement.	Peer-led labs, iterative feedback, and inclusive evaluation.	T3	Maintain ongoing user feedback, high impact even on low budgets.
Inclusive teaching and pedagogy.	Universal Design for Learning, neurodiversity strategies, micro-credentials.	T4	Micro-credentials sustain educator learning, and teacher buy-in is critical.
Workplace hiring and onboarding.	Accessible e-recruitment, explicit materials, and disability-aware onboarding buddies.	T4	Train hiring managers, improves applicant diversity and experience.
Cultural, tourism, and heritage training.	Accessible tours, VR/AR with accessible controls; staff sensitivity workshops.	T4	Provide an adaptive tour training, build local partnerships.
Emergency preparedness and security.	Inclusive drills, facilities, cross-training, and accessible evacuation planning.	T3	Wheelchair evacuation is often overlooked, treat preparedness as routine, not exceptional.

Table 1 Theme-derived synthesis of cross-sector training needs and competencies (grounded in participants' accounts and aligned with Themes T1–T4).

Notes: Abbreviations defined in text (WCAG, EN 301 549, EAA/Directive 2019/882).

4.1 EMPATHY AS ORGANIZATIONAL COMPETENCE: LINKING T1 TO PRIOR LITERATURE

Empathy moves from sentiment to proceduralized practice through user-led activities, carefully curated evidence, and concrete design decisions. Teams make better choices when empathy is institutionalized as a competence embedded in methods, artifacts, and routines (Drouet et al. 2023). Accessibility education similarly argues for cross-role pedagogy, mixed audience training, and champions networks that connect academia and workplaces (Coverdale et al. 2022).

A practical path is to involve non design stakeholders so empathy work travels beyond UX teams; pair awareness methods with measurable follow through such as edits to copy, interaction flows, and acceptance criteria; sustain participation through user panels, partnerships with Organizations of Persons with Disabilities, and regular feedback loops; and invest in role sensitive training and internal champions so practices persist across turnover and projects.

Participants' emphasis on empathy, user involvement, and iterative testing aligns with a dynamic diversity view that treats variability as a design premise rather than a late exception (Gregor et al. 2002). Training that makes diversity visible in concrete tasks, paired with participation and reflection, counters implicit ableism (Goodley 2014). Interviewees consistently highlighted barrier-focused, user-led activities and direct engagement with people with disabilities as routes from abstract sympathy to practical understanding. Some mentioned role-play or simulations, but only when co-designed, critically examined, and followed by specific changes. Legal and policy literacy operates alongside empathy as a baseline across digital, built, and service environments. Practitioners need working knowledge of WCAG 2.2, EN 301 549, and Directive (EU) 2019/882 for digital services, and EN 17210 with relevant national building rules

for the built environment, so that requirements become acceptance criteria and verification methods. Participants stressed the need to clarify jurisdictional overlaps, agree on explicit Definition of Done criteria, and recognize that legal compliance is the minimum standard. Calls for formal certification point to the need to professionalize accessibility, as with other regulated competencies.

AI and emerging technologies add competencies. Interviewees recommended integrating AI ethics, bias mitigation, dataset provenance, and practical tool literacy into role-specific development for managers, designers, developers, and agile facilitators. Live demonstrations of the limits of automated captioning or alt text generation were described as effective. Deployment should wait until fundamentals are in place, and automation should not replace user involvement or expert review.

Digital transformation requires collaboration skills. Cross-functional training, shared checkpoints during agile ceremonies, and partnership building were consistently linked to higher-quality outcomes. Teams that plan for accessibility from project kickoff rely less on late remediation and maintain continuous user testing with peer-led labs and rapid feedback loops, even in low-budget contexts. Sector-specific capacity is also required. Education benefits from micro credentials, mentorship, and Universal Design for Learning. Employment benefits from accessible hiring platforms, manager training, and disability aware onboarding buddies. Cultural and tourism settings benefit from staff-sensitivity workshops, accessible tours, and the careful use of VR or AR with accessible controls. Emergency preparedness and cross-training of facilities are essential, especially in legacy buildings. To make safety actionable, track simple indicators such as the percentage of facilities with an inclusive drill schedule, the percentage of staff trained on accessible evacuation, and the median time to assist a wheelchair user during drills.

4.2 CHANGE MANAGEMENT AND INSTITUTIONAL TRANSFORMATION: LINKING T2 TO PRIOR WORK

Participants emphasized leadership and change management. Efforts that depend on a few motivated individuals rarely persist when resources tighten or priorities shift. When leaders model inclusive behavior, assign ownership, and allocate budget, accessibility becomes routine. Interviewees described mechanisms that sustain change: steering groups with a clear mandate, agreed acceptance criteria, and regular reporting that includes user-centered metrics. Normalizing experimentation, with permission to fail and learn openly, signals an institutional stance toward learning rather than compliance alone.

Governance was central to participants' accounts, and many noted persistent uncertainties about ownership. Clear responsibility chains reduce diffusion of responsibility and help teams integrate accessibility into procurement, product management, IT operations, communications, and HR. Participants proposed practical devices: a central steering group with representation from affected units, a lightweight accountability framework with a short set of key indicators, and a regular cadence for reviewing progress alongside user feedback. These practices embed accessibility into organizational infrastructure rather than relying on personal goodwill.

Calls for clear ownership, a shared Definition of Done, and visible accountability respond to Shakespeare's critique of rhetoric without rights realization and echo Roulstone's view that technology policy requires organizational governance to prevent innovation from reproducing exclusion (Roulstone 2016; Shakespeare 2006).

4.3 ANTICIPATING POLICY SHIFTS: THE EAA AND BEYOND

Directive (EU) 2019/882 serves as both a motivator and a stressor. It motivates by setting a clear floor and creating urgency. It stresses budgeting, procurement, vendor management, and legacy systems. Participants asked for sector-specific interpretations and templates that translate the Act into procurement language, including Definition of Done clauses, verification plans, and vendor criteria. They also warned that compliance can drift from inclusion unless shaped by user feedback and iterative testing. A policy lens combined with a learning lens is most productive. Regulation creates alignment and budgets, while sociocultural learning keeps implementation user-centered and adaptive over time (EU 2019).

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Beyond the Act, participants described a broader trajectory. Public organizations face rising expectations for transparency in accessibility performance, while tourism and cultural institutions compete on inclusive reputation. Sector work shows that destinations can brand around accessibility, audit routes and points of interest, and publish accessibility information as both policy implementation and market signaling (Darcy and Buhalis 2011). For digital services, WCAG 2.2 provides measurable targets that can be embedded in procurement, design systems, and release criteria (Henry 2023).

This trajectory aligns with rights-based and capability approaches. Organizations should avoid treating legislation as a ceiling and use policy to expand people's freedoms to participate. Standards and guidelines become levers for redesigning processes, so participation is routine rather than exceptional.

4.4 OPERATIONALIZING ACCESSIBILITY, FROM STRATEGY TO PRACTICE: LINKING T3 AND T4

In this discussion, examples from adjacent contexts are interpreted as extensions of organizational routines, training needs, and governance practices, not as claims to exhaustive coverage of those domains. Participants outlined practical strategies. In cultural institutions and libraries, VR or AR can widen access when paired with accessible controls, captions, transcripts, and tactile alternatives. Self-audit tools and centralized knowledge portals support continuity when feedback loops, mentoring, and role-specific guidance accompany them. These practices show how accessibility is sustained through coordinated organizational routines rather than isolated technical fixes.

Two issues need attention. First, treat emergency preparedness as part of universal design. Facilities and security teams need training for inclusive evacuations and drills, especially in legacy buildings. Second, treat AI as assistive and subject to human review. Dataset provenance, bias checks, and user testing with people with disabilities are prerequisites for deployment.

These details convert policy intentions into lived accessibility. Co-design, feedback loops, and incremental testing implement the HCI insight that systems must accommodate changing needs across contexts (Gregor et al. 2002). Governance for AI and digital services is required to ensure innovation widens participation rather than narrows it (Roulstone 2016).

5 CONTRIBUTIONS

This study advances accessibility research in three ways. First, it frames accessibility as organizational learning produced through social interaction, shared tools, and guided participation across sectors, thereby extending work that often treats implementation within single domains. Second, it shows how empathy becomes operational competence through governance routines, co-design, role-specific training, and accountability mechanisms. Third, it demonstrates that accessibility is sustained through the interaction of standards, collaboration, and organizational logistics in everyday practice, refining accounts that emphasize compliance or design in isolation.

5.1 IMPLICATIONS FOR PRACTICE AND POLICY

For practitioners, take a sequenced, integrated approach. Start with a shared-mindset curriculum that includes learning with people with lived experience of disabling barriers, policy literacy tied to concrete delivery checkpoints, and role-specific re-possibilities. Build this into procurement, design systems, quality assurance, onboarding, and continuing professional development so accessibility becomes routine organizational practice rather than a specialist add-on.

For policy and leadership, use implementable templates aligning procurement, vendor management, and service contracts with Directive (EU) 2019/882 (EU 2019). Include clauses for accessibility acceptance criteria, verification plans, and remediation timelines. Signal priority through budget lines, recognition in performance reviews, and public reporting that combines compliance and user experience measures.

In simulations, take a balanced position. They can raise awareness when co-designed, caveated, and followed by action, but must not replace engagement with lived experience through Organizations of Persons with Disabilities, user panels, and task-based testing.

5.2 LIMITATIONS AND TRANSFERABILITY

Transferability is supported through thick description of roles, sectors, and the organizational mechanisms participants described (e.g., governance ownership, Definition of Done/acceptance criteria, and learning infrastructures). Readers can assess relevance to comparable contexts, particularly organizations implementing Directive (EU) 2019/882 and harmonized standards, by considering similarities in service type, maturity, resourcing, and regulatory obligations.

Several limitations should be considered when interpreting these findings. First, the study is based on a purposive, cross-sector sample (N = 16) designed for depth rather than statistical generalization. Second, one collaborating organization (A1) provided interviewer-prepared summaries rather than full transcripts, which reduces granularity; we therefore treated these materials as lower-resolution data and relied primarily on verbatim transcripts for fine-grained interpretive claims. Third, the dyadic group discussion provides interactional insight but lacks the breadth of dynamics typical of larger focus groups.

6 CONCLUSIONS

In relation to our research question, the findings show that professionals conceptualize, implement, and advocate for accessibility as a holistic organizational practice rather than merely as compliance. Empathy and cultural competence provide the ethical foundation; governance and policy translate values into accountable routines; technology and collaboration shape implementation; and sustained learning supports long-term change. The study, therefore, contributes an empirically grounded account of how accessibility is translated from strategy into everyday work across sectors, while remaining limited to professional perspectives rather than direct evaluation of compliance outcomes.

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COMPETING INTERESTS

The authors have no competing interests to declare.

AUTHOR AFFILIATIONS

Ali Mohamed orcid.org/0000-0002-8560-7156
Dalarna University, SE

Jan Gulliksen orcid.org/0000-0002-2411-6417
KTH Royal Institute of Technology, SE

Sofia Strömqvist orcid.org/0009-0002-3635-7741
KTH Royal Institute of Technology, SE

Masood Rangraz orcid.org/0000-0003-3506-2327
KTH Royal Institute of Technology, SE

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